

How has been the acceptability in other countries?

Japan was the first country to test and implement 3G. The acceptability has been good for 3G there, with internet access, gaming and music download being some of the prominent applications. The objective of launching 3G in Japan and Korea was to develop an internal communication network and hence the access charges for service providers were kept low, resulting in low 3G charges on consumers, and hence greater popularity and acceptance. 3G has done well in most countries due to its good features. It has been successfully launched and is being operated in more than 50 countries currently.

On the other hand, the financial performance of 3G operators has been poor in some regions due to high debts on service providers or high-access charges; especially in Europe. The actual performance of 3G in most places has been lower than the specified speed standards and has caused dissatisfaction among consumers, leading to lower acceptance rates for the service.

What are the major concerns involving 3G?

3G rollout in many countries has resulted in factors hindering acceptance of service among consumers and profitability of service providers; India would do well to learn from them. Additionally, we foresee certain issues the operators are likely to face in India:

- High revenue expectations from 3G services resulted in high input fee/access charges levied on service providers for 3G service licenses.
- The price of 3G mobile services (including internet access) was kept excessively high in some countries, especially Europe, reducing the attractiveness of the 3G offering.
- Heavy upfront capital investment is required for developing local infrastructure capable of supporting 3G services in any country.
- 3G phones are more expensive compared to 2G and other lower-end phones, making the switch to 3G financially unfavourable for common consumers, especially in India.
- In India a substantial proportion of the population does not require 3G voice and data services in hand-held devices; this would limit the spread of 3G in the short term.
- Health impact of electromagnetic waves would always remain a cause for concern.
- There is a general lack of awareness about 3G services even among the educated population in India. It would take time for 3G services to pick up, as has been the case globally.

What's ahead for 3G?

Organizations worldwide are involved in a race to develop the next level of wireless communi-

digit

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cation, and hence to grab and maintain a substantial market share. Some of the possible new areas of communication standard evolution can be:

- **Universal networking:** development of architectures and middleware that support universal networking between all communication devices leading to smooth communication and data/application transfer from one device to another.
- **Customized services:** the user would have the power over which function can be served by which device (one device acting as the central controller for all other devices the user owns, music being played on one device while video on another, etc.)
- **Higher network speeds:** with better bandwidth utilization and multiplexing technologies, network speeds are expected to go up, enabling on-the-fly data transfer from one device to another.
- **Other auxiliary services:** other features and services, such as heightened network security through Bluetooth device-based protection, identification services, information services customized to user preferences, centralized billing, etc would come up with the evolution of communication technology.
- **4G:** An upgrade over the current 3G system involving some advanced features, such as better spectrum utilization, greater network speed, more simultaneous users per cell phone, better inter-operability across heterogeneous networks, etc.

WiMAX

Launch of WiMAX in India

The stage is set for the launch of WiMAX in India with the finalisation of dates for the auction of the 2.5 / 3 GHz spectrum by the Department of Telecommunication (DoT). The 3G and WiMAX spectrums are scheduled to be auctioned during January and February 2009.

The timetable	
Date	Action
Dec 11 & 13 2008	Bidders meet
Dec 19 2008	Pre-bid conference
Dec 31 2008	Last date of submitting applications
Jan 05 2009	Pre-qualification of bidders
Jan 16 2009	First phase of auctioning
Feb 09 2009	Second phase of auctioning

In a recent announcement, the India chapter of the WiMAX Forum has made public its plans to set up an applications lab at the Indian Institute of Technology (IIT), Delhi. In view of the high business potential of the Indian market, the WiMAX community places high importance on India; this is evident from the plan to set up this lab, only

the third of its kind across the globe after USA and Taiwan.

According to the WiMAX forum, the market for WiMAX in India, which includes device and service sales, will reach \$13 billion by the year 2012. The forum pegs the number of WiMAX users by 2012 at 27.5 million, an approximate 20 per cent share of the world's total WiMAX user base at that time. According to independent research by Maravedis, a market research firm, the subscriber base for WiMAX in India is expected to reach 21 million by the year 2014.

Leading global WiMAX companies, such as Alcatel-Lucent, Motorola and Samsung, are starting their operations in India. Nokia has announced the launch of its first ever WiMAX-enabled phone which is scheduled to be launched in India.

What is WiMAX?

WiMAX, or "Worldwide Interoperability for Microwave Access", is a broadband wireless point-to-multipoint specification from the IEEE 802.16 working group, enabling wireless broadband networks with high speed. Leaps and bounds ahead of WiFi, WiMAX is designed to operate as a wireless metropolitan-area network.

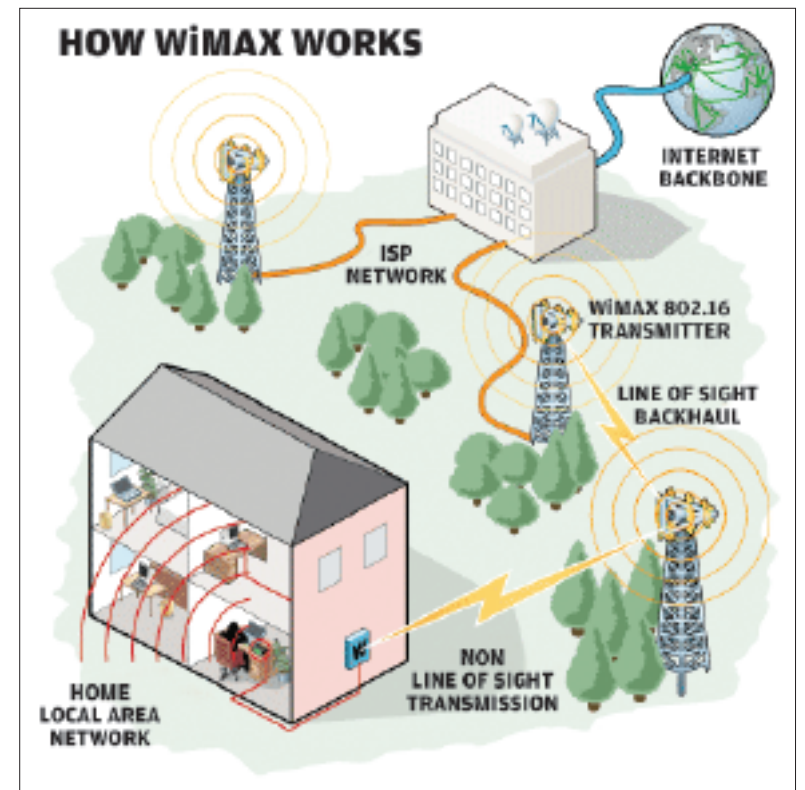
WiMAX promises faster and superior performance, adaptability to various applications, advanced IP-based architecture (enabling integration of various applications with the internet) and reduced rates to consumers.

WiMAX specifications	
Parameter	Value
Range: Line-of-sight (LOS)	~50 Km
Range: Non-line-of-sight (NLOS)	~10 Km
Maximum data speed	~70 Mbps
Licensed frequency band	2-11 GHz
Un-licensed frequency band	10-66 GHz
Switching	Packet
Multiplexing	SOFDMA
Modulation	"BPSK, QPSK, 16QAM, 64QAM"

Source: <http://grouper.ieee.org/groups/802/16/>

WiMAX supports a range of modulation algorithms to facilitate the realization of optimal bandwidth in all conditions. The theoretical maximum data bandwidth of 75Mbps can be achieved with WiMAX using 64QAM 3/4 modulation (only under optimal transmission conditions).

Fixed WiMAX is limited to providing wireless network access to fixed devices, a limitation eradicated by mobile WiMAX. WiMAX does not have the need for a direct line of sight between the source and endpoint of communication/data transfer, unlike most other wireless technologies. Mobile WiMAX supports SOFDMA



Evolution of WiMAX	
Specification	Description
802.16-2001	LOS access in 10-66 GHz spectrum
802.16c-2002	Improved QoS; support for HIPERMAN
802.16a-2003	Added 2-10 GHz to existing standards
802.16d-2004	Fixed WiMAX; supported OFDMA 256
802.16e-2005	Mobile WiMAX; enhanced OFCDMA multiplexing
802.16f-2006	Information base management
802.16g-2007	Mobility management
HIPERMAN	Equivalent of 802.16 in Europe
WiBro	Equivalent of 802.16 in Korea

(Scalable Orthogonal Frequency Division Multiple Access), an enhancement over OFDMA256 which facilitates non-line-of-sight communication. SOFDMA also brings in better throughput and minimized sound fading in multiple path scenarios.

What are the key highlights of WiMAX?

WiMAX has some important features that have attracted the attention of the whole information and communication community ranging from internet operators and mobile telecom operators to fixed line telecom service providers, and so on. The attraction of possible gains in the future has brought many technology firms into the WiMAX fold:

- **IP-based network:** WiMAX specifications are developed over IP and support IP-based applications. The need to unify various communication networks and to cut costs has inspired